

| UEC522: DATA COMMUNICATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |   |   |     |
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| <p><b>Course Objective:</b> To introduce basic concepts of Data communication with different models. Enumerate the physical layer, Data Link Layer, Network Layer, Transport Layer and Application Layer, explanation of the function(s) of each layer. Understanding of switching concept and different types of switching techniques.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |   |     |
| <p><b>Syllabus</b></p> <p><b>Overview of Data Communication and Networking:</b> Data communications, Networks, The Internet, Protocols and standards, Layered tasks, OSI model, TCP /IP protocol Architecture.</p> <p><b>Physical layer:</b> Analog and digital, Analog signals, Digital signals, Analog versus digital, Data rate limit, Transmission impairments, Line coding, Block coding, Sampling, Transmission mode, Modulation of digital data, Telephone modems, Modulation of analog signal, FDM, WDM, TDM, Guided media, Unguided media, Circuit switching, Telephone networks, DSL technology, Cable modem.</p> <p><b>Data link layer:</b> Types of errors, Detection, Error correction, Flow and error control, Stop and wait ARQ, go back n ARQ, Selective repeat ARQ, HDLC, point to point protocol, PPP stack, Random access, Controlled access, Channelization, 802.15.4; FDDI, Bluetooth; Telephone networks, DSL technology, Cable modem, SONET/SDH. Connecting devices, Backbone network, Virtual LAN, Cellular telephony, Satellite networks.</p> <p><b>Network layer:</b> Internetworks, Addressing, Subnetting, Routing, ARP, IP, ICMP, IPV6, Unicast routing, Unicast routing protocol, Multicast routing, Multicast routing protocols.</p> <p><b>Transport layer:</b> Process to process delivery, User datagram protocol (UDP), Transmission control protocol (TCP), Data traffic, Congestion, Congestion control, Quality of service, Techniques to improve QOS, Integrated services, Differentiated services, QOS in switched networks.</p> <p><b>Application layer:</b> Client server model, Socket interface, Name space, Domain name space, Distribution of name space, DNS in the internet, Resolution, DNS messages, DDNS, Encapsulation, Electronic mail, File transfer, HTTP, World wide web (WWW), Digitizing audio and video, Audio and video compression, streaming stored audio/video, Streaming live audio/video, Real time interactive audio/video, Voice over IP.</p> <p><b>Switching:</b> Introduction to Virtual circuit switching including frame relay, ATM and Softswitch Architecture, Packet switching, Message Switching.</p> <p><b>Familiarization with standards:</b> <i>IEEE 802.3 to 802.6 and 802.11, IEEE/ISO/IEC 8802-15-6-2017, IEEE 802.1CS-2020.</i></p> |   |   |   |     |
| <p><b>Course Learning Objectives (CLO)</b></p> <p>The students will be able to:</p> <ol style="list-style-type: none"> <li>1. Understand the layered architecture of Internet's reference models: OSI &amp; TCP/IP and basis of physical layer and media.</li> <li>2. Analyze the design issues, framing, error detection and correction, channel allocation techniques and link layer protocols.</li> <li>3. Evaluate the network layer, internetworking, IP and IP addressing mechanism.</li> <li>4. Apply various routing algorithms, elements of transport protocols, congestion control, and QOS in networks.</li> <li>5. Understand various communication applications like email, web browser, and familiarization with cryptography and network security.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |   |   |     |

**Text Books**

1. Ferouzan, Behrouz A., *Data Communications and Networking with TCP/IP Protocol Suite*, TATA McGraw Hill (2022) 6th Edition.
2. Tanenbaum, Andrew S., *Computer Networks*, PHI (2013) 5th Edition.

**Reference Books**

1. *Microwave & RF Design*, Michael Steer, Sci.Tech Publishing, 2009.
2. Stallings William, *Data and Computer Communication*, Pearson Education (2017) 10th Edition.
3. James F. Kurose, *Computer networking: A top-down approach*, Pearson Education (2017), 6th Edition.